Comparative Analysis of Input Modalities for 3D CAD Generation using Multimodal Large Language Models

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Computer-Aided Design (CAD) is a core technology in modern product design and manufacturing processes. However, creating precise geometric models that reflect a user's design intent requires specialized knowledge and repeated modeling experience. With the recent development of multimodal large language models (LLMs), AI-based CAD generation using natural language and image inputs has been actively studied. Previous research quantitatively compared text-, image-, and vector-based input methods for 2D planar shapes and showed that more structured input information improves generation accuracy and stability. This study extends the existing input modality comparison framework to 3D CAD generation and investigates whether the findings observed in 2D shape generation remain valid for complex 3D solid models. To this end, 3D CAD models of mechanical parts with different complexity levels are generated using natural language descriptions, 2D images, and structured vector data under the same generation pipeline. Since 3D modeling requires additional spatial information such as depth, cross-sectional structure, and geometric relationships, this study analyzes the effect of modality-specific information density on generation performance. The results provide practical guidelines for optimal input design in LLM-based 3D CAD automation systems and suggest that natural language interfaces can support early-stage product design.

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